Jeison Mesa, Curriculum Vitae

Statistician, data scientist and AI engineer with 10+ years of experience in R and Python, from climate research to industry machine learning and LLM agents.

Summary

I'm a statistician, data scientist and AI engineer with more than ten years of experience, working mostly in R and Python. I started in climate and agricultural research, where I learned to build models that other people had to trust and act on. Since 2021 I've applied the same habits to industry problems: customer churn and segmentation, pricing, forecasting, recommendation and text embeddings, and more recently LLM-based agents.

I like the part of the job between the notebook and production: cleaning up the data, making the model reproducible, and getting it running somewhere useful (AWS SageMaker, GCP Vertex AI, Kubeflow, Docker).

Experience

AI Engineer · Globant, on assignment at Disney

Dec 2025 – Present · Remote, Colombia

Building LLM-based systems for Disney:

- A multi-agent system that reads a codebase and generates its documentation, with structured context injection, schema validation between agents and a clear split between deterministic and LLM steps to make results reliable and explainable.
- A conversational intake agent that standardizes internal product requests, pushes back on vague input and flags overlap across teams.
- An automated reporting pipeline that gathers data from project-management tools into one reporting model and produces scheduled executive and team reports, with a dry-run step before anything is distributed.

Tools: Python, LangGraph, Azure OpenAI, REST APIs, Docker, Git, CI/CD.

Senior Data Scientist · Globant

May 2021 – Dec 2025 · Remote, Colombia

Sportian account (Aug 2023 – Dec 2025)

- Built statistical and machine learning models for sports-industry products, mainly pricing and match analytics.
- Worked with time series, econometric methods and embeddings on historical sales and event data.

Augmented Staffing (Feb 2022 – Jan 2023)

- Forecasting and clustering of internal staffing tickets, used to plan recruiting.
- Built a semantic search and recommendation tool that matched candidates to tickets using text embeddings, plus a seniority estimation model in PyTorch with an ordinal loss.
- Ran the pipelines on Kubeflow.

DirecTV (May 2021 – Dec 2021)

- Customer segmentation for the DirecTV GO subscriber base.
- Churn model that combined behavioural usage data with sentiment from text, plus an analysis of subscription pricing and a market basket analysis to guide retention campaigns.
- Pipelines on AWS SageMaker.

Professor (part-time) · Universidad del Valle and Universidad Católica

Aug 2020 – Jun 2021

Taught time series analysis and probability at Universidad del Valle, and inference and probability at Universidad Católica.

Research Assistant and Data Scientist · CIAT (International Center for Tropical Agriculture)

2014 – 2020 · Cali, Colombia

Climate and agriculture group. I automated, in R, the full pipeline from climate data to crop simulation, and modelled climate variables and crop yields across Latin America and beyond. Highlights:

- Impacts of near-term climate change on major crops in Latin America and the Caribbean (for the IDB and USAID), using DSSAT and ORYZA crop models together with machine learning and geostatistics.
- Identification of "climate service hotspots" worldwide, where agro-climatic information could help close yield gaps (AgMetGaps).
- Statistical models behind the AClimate Colombia seasonal forecast platform (paper).
- Regional climate change plan for the Orinoco region (Meta, Casanare, Vichada and Arauca): spatial analysis, climate clusters and yield simulation for the main production systems.
- Vulnerability analysis and classification of production systems for Colombia's Fondo Adaptación, with CORPOICA.

Earlier work, 2010 – 2014

- **Consultant, Universidad del Valle** (2012–2013): statistical design and analysis for two engineering theses, one on constructed wetlands (factorial and repeated-measures designs, mixed models) and one on equations of state (non-linear models, bootstrap).
- **Professor, Universidad Santiago de Cali** (2013): a course on statistics in R for research faculty.
- **Statistical Assistant, Universidad del Valle** (2010–2013): analysis for the quality and accreditation office of the health faculty, with Excel macros and R automation.

Education

- **B.Sc. in Statistics**, Universidad del Valle, Cali (2008 – 2014). Thesis: *Comparison of estimation methods for the fractional differencing parameter in conditionally heteroscedastic long-memory time series.*

Skills

Languages	R (advanced), Python, SQL
Modelling	Regression and mixed models, clustering, dimensionality reduction (PCA, kernel PCA, t-SNE, UMAP), gradient boosting, deep learning (PyTorch), anomaly detection
Time series and spatial	ARIMA, GARCH, state-space models and Kalman filters, Gaussian processes, geostatistics, raster and satellite data
Text and GenAI	LLM agents and multi-agent systems (LangGraph, LangChain), prompt engineering, embeddings and vector search, topic models, Hugging Face, OpenAI and Azure OpenAI
Data and cloud	PySpark, sparklyr, Polars and pandas, Parquet, Databricks, AWS (SageMaker, Athena, S3), GCP (Vertex AI), some Azure
Engineering	Git and GitHub, Docker, Kubeflow, Linux and WSL, DVC
Domain	Climate impact modelling, crop simulation (DSSAT, ORYZA), pricing, churn and customer segmentation
Communication	Quarto, R Markdown, LaTeX

Languages spoken: Spanish (native), English (intermediate).

Selected publications

Full list on Google Scholar.

- Valencia JB, Guryanov VV, **Mesa-Diez J**, Diaz N, Escobar-Carbonari D, Gusarov AV (2024). Predictive assessment of climate change impact on water yield in the Meta River Basin, Colombia: an InVEST model application. *Hydrology*.

- Valencia JB, Guryanov VV, **Mesa-Diez J**, Tapasco J, Gusarov AV (2023). Assessing the effectiveness of the use of the InVEST annual water yield model for the rivers of Colombia: a case study of the Meta River Basin. *Water*.
- Ordoñez L, Vallejo E, Amariles D, **Mesa J**, Esquivel A, Llanos-Herrera L, et al. (2022). Applying agroclimatic seasonal forecasts to improve rainfed maize agronomic management in Colombia. *Climate Services*, 28, 100333.
- Ochoa-Muñoz AF, Peña-Torres JA, García-Bermúdez CE, Mosquera-Muñoz KF, **Mesa-Diez J** (2022). On characterization of sensory data in presence of missing values: the case of sensory coffee quality assessment. *Ingeniare*, 30(3), 564–573.
- Valencia JB, **Mesa J**, León JG, Madriñán S, Cortés AJ (2020). Climate vulnerability assessment of the *Espeletia* complex on páramo sky islands in the northern Andes. *Frontiers in Ecology and Evolution*, 8, 565708.
- Sotelo S, Guevara E, Llanos-Herrera L, et al., **Mesa J**, et al. (2020). Pronósticos AClimateColombia: a system for the provision of information for climate risk reduction in Colombia. *Computers and Electronics in Agriculture*, 174, 105486.
- Mwongera C, Ramirez-Villegas J, Mutua J, Mora B, **Mesa J**, et al. (2020). *Climate vulnerability assessment for selected value chain commodities in Zambia*. Alliance of Bioversity International and CIAT.
- Rodríguez De Luque JJ, González-Rodríguez CE, Gourdji S, Mason-D'Croz D, Obando-Bonilla D, **Mesa-Diez J**, Prager SD (2016). Socio-economic impacts of climate change in Latin America and the Caribbean: 2020–2045. *Cuadernos de Desarrollo Rural*, 13(78), 11–34.
- Gourdji SM, **Mesa-Diez J**, Obando-Bonilla D, Moreno-Cadena LP, Navarro-Racines CE, Fisher M, Prager SD, Ramirez-Villegas J. *Climate change vulnerability in the agricultural sector in Latin America and the Caribbean.* CORE.

Talks and conferences

- *Simulated near-term climate change impacts on major crops across Latin America and the Caribbean.* AGU Fall Meeting, San Francisco, 2016 (co-author, abstract).
- *Enabling the breeding of climate-smart crops* and *Climate services for agriculture: forecast skill no guarantee of success?* Global Climate-Smart Agriculture Conference, Bali, 2019 (co-author, abstract book).
- *Servicios agro-climáticos para la agricultura: ¿sueño o realidad?* Congreso SOCOLEN, Cali, 2018 (co-author).
- *Comparación de métodos de estimación para el parámetro de diferenciación fraccional en series de tiempo de memoria larga condicionalmente heterocedásticas.* Jornadas en Estadística e Informática, ESPOL, Guayaquil, 2014, and Escuela de Verano, Universidad Nacional de Colombia (Medellín) and Universidad del Valle, 2013 (speaker).

Awards

- **UNFCCC Momentum for Change award**, 2017, as part of the CIAT climate services team.
- **Best Research Outcome**, CIAT, 2017.
- Academic stimulus, Universidad del Valle: second best position (2010), fourth best position (2011).

- Best graduate, Colegio César Conto, 2007.

Selected courses

- Quantitative Ecology and Conservation of Biodiversity, and Selection Analysis of Animal Resources Using R. CEBIO, Lima, 2014.
- Classification Methods: Grouping and Discriminant Analysis. ESPOL, Guayaquil, 2014.
- Spatial Data Analysis, Non-linear Models, Econometrics. Universidad del Valle, 2012–2013.